

1. What is a Calculated Field in Tableau? How is it different from a Regular Field?

Ans:

A **Calculated Field** in Tableau is a new field created by applying a formula or expression to existing fields in the dataset. It allows users to perform calculations, create business metrics, and derive new insights without modifying the original data source.

Example:

Suppose a dataset contains:

- Sales
- Profit

A calculated field can be created as:

`[Profit Ratio] = [Profit] / [Sales]`

This creates a new field called **Profit Ratio**.

Difference Between Calculated Field and Regular Field

Regular Field	Calculated Field
Comes directly from the data source	Created by the user in Tableau
Stores original data	Stores results of calculations
Cannot perform custom logic by itself	Can apply formulas, conditions, and aggregations
Example: Sales, Customer Name	Example: Profit Ratio, Discount Category

2. Explain the Difference Between Row-Level and Aggregate-Level Calculations with Examples.

Ans:

Row-Level Calculation

A row-level calculation is performed **for each individual record (row)** in the dataset before aggregation occurs.

Example:

`[Profit] - [Cost]`

Dataset:

Product Profit Cost

A	100	70
B	150	90

Result:

Product Profit Margin

A	30
B	60

The calculation is applied to every row separately.

Aggregate-Level Calculation

An aggregate calculation is performed on summarized data using functions such as:

- SUM()
- AVG()
- MIN()
- MAX()
- COUNT()

Example:

SUM([Profit]) / SUM([Sales])

Dataset:

Product Profit Sales

A 100 500

B 150 600

Calculation:

$(100 + 150) / (500 + 600)$

= 250 / 1100

= 22.73%

The calculation is performed on aggregated values rather than individual rows.

Difference Summary

Row-Level Calculation

Evaluated for each row

No aggregation function required

Produces row-wise results

Example: [Sales]-[Cost]

Aggregate-Level Calculation

Evaluated on summarized data

Uses SUM(), AVG(), COUNT(), etc.

Produces a single summarized result

Example: SUM([Sales])

3. How Does Tableau Handle Null Values in Calculated Fields? Illustrate with an Example.

Ans:

A **Null** value represents missing or unknown data. Tableau generally returns **Null** when a calculation involves a Null value unless special handling is applied.

Example 1: Without Null Handling

Dataset:

Customer Sales

A 1000

B Null

Calculated Field:

[Sales] * 2

Result:

Customer Result

A 2000

B Null

Since the value is missing, Tableau returns Null.

Example 2: Using IFNULL()

To replace Null values:

IFNULL([Sales], 0)

Result:

Customer Result

A 1000

B 0

Example 3: Using ZN()

For numeric fields:

ZN([Sales])

Result:

Customer Result

A 1000

B 0

Common Functions for Handling Null Values

Function

Purpose

IFNULL(expression,value) Replaces Null with specified value

ISNULL(expression) Checks whether a value is Null

ZN(expression) Replaces numeric Null values with 0

Example Using ISNULL()

IF ISNULL([Sales]) THEN

"Missing Data"

ELSE

"Available"

END